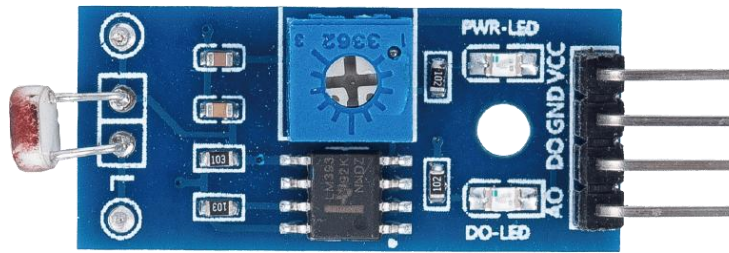


Photoresistor Module



The Photoresistor module can detect light intensity in the environment and be used to adjust device brightness, detect day and night, or activate a light switch. The module supports analog output and digital output, equipped with a power indicator light, a signal indicator light, and a potentiometer for adjusting the light intensity threshold.

Feature

- Includes an LED indicating the relationship between the light intensity and the threshold.
- Includes a potentiometer to adjust the sensitivity of the digital output. This can be used to set a threshold so that when the ambient light intensity exceeds this value, the module outputs a low level; otherwise, it remains high.
- Equipped with a wide voltage LM393 comparator.

Specification

- Supply Voltage: 3.3V - 5V
- PCB Size: 32 x 14mm
- Output Signal Type: DO and AO

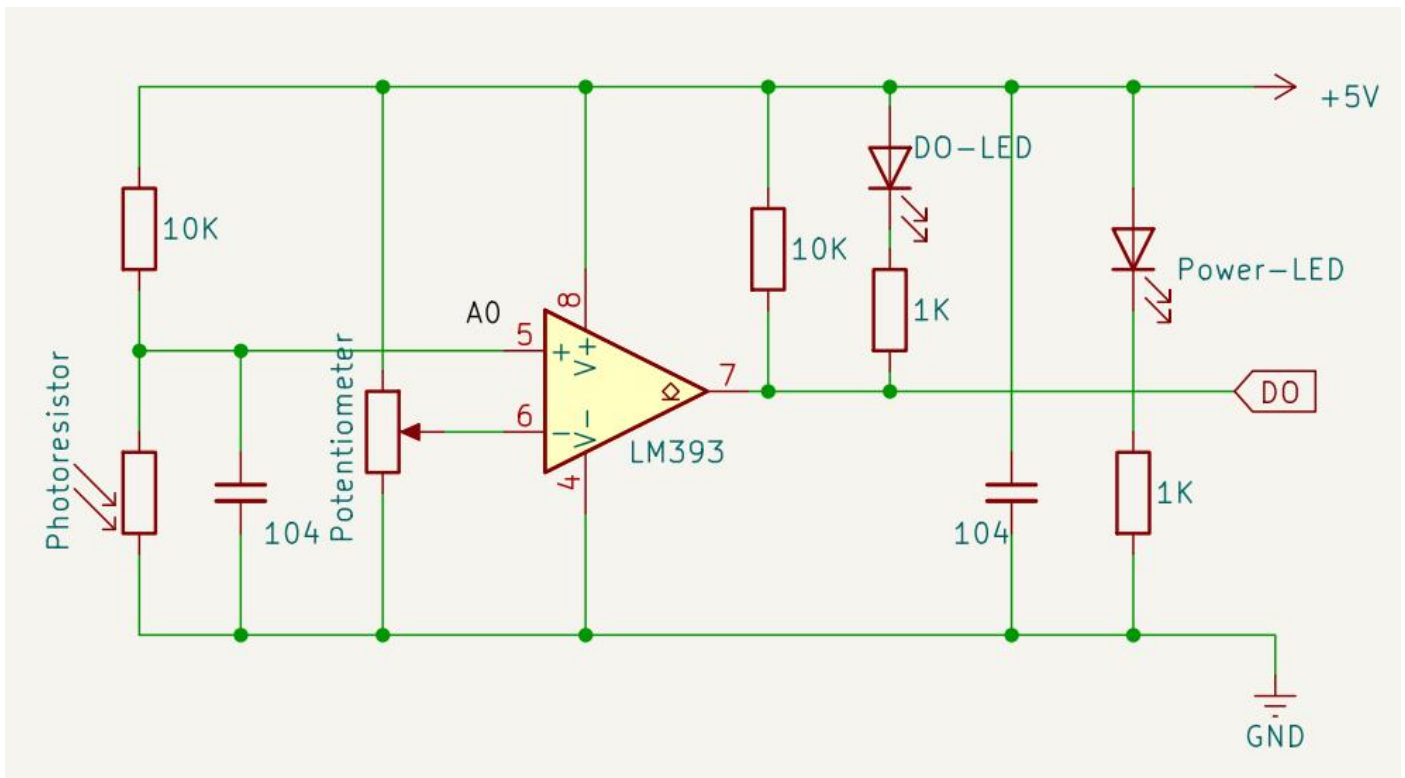
Pinout

- **VCC:** This is the positive power supply input from the main control.
- **GND:** Ground connection.
- **DO:** Digital output. When the intensity of the light exceeds the threshold value (set by the potentiometer), DO becomes LOW; otherwise, it remains HIGH.
- **AO:** Analog output. The stronger the light, the lower the output value; conversely, the weaker the light, the higher the output value.

Tip

- ◆ The photoresistor module is sensitive to ambient light and is generally used to detect the brightness of the surrounding environment, triggering microcontrollers or relay modules.
- ◆ When the ambient light intensity is below the set threshold, the DO pin outputs a high level. When the ambient light intensity exceeds the set threshold, the DO pin outputs a low level.

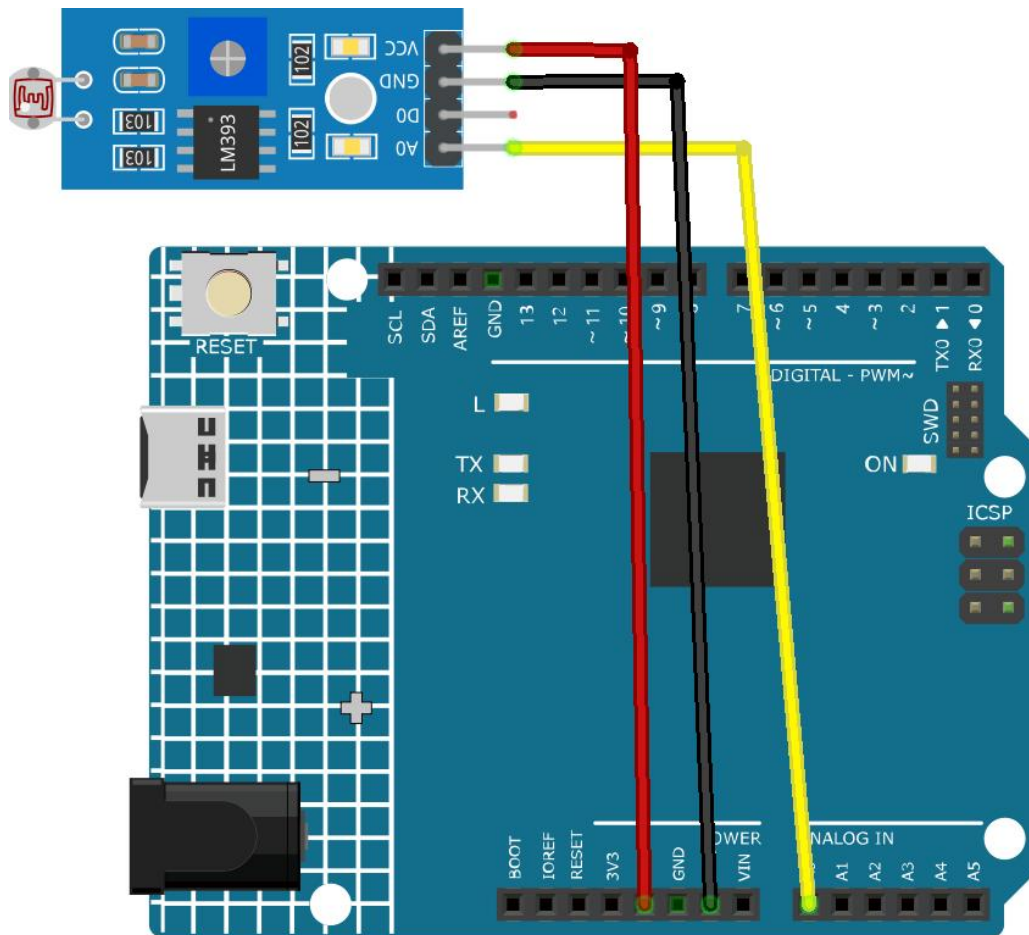
Schematic Diagram



How to Use in Arduino?

Build the Circuit

Photoresistor Module	Arduino Board
VCC	5V
GND	GND
DO	
AO	A0



Copy and Upload the Code

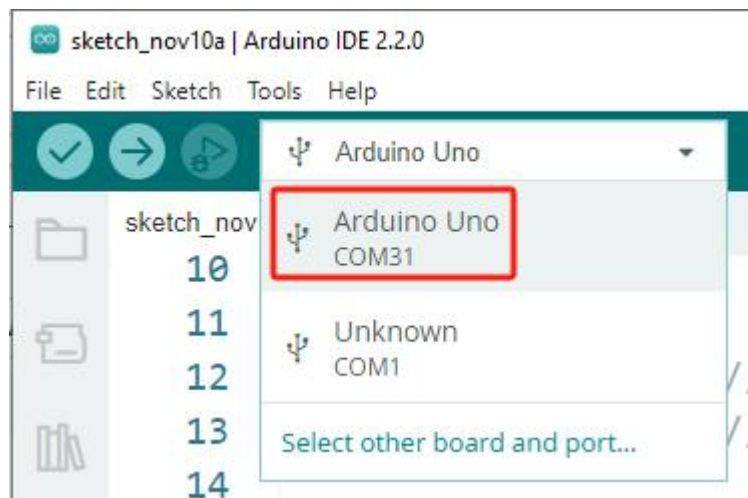
1. Copy and paste the following code to the Arduino IDE.

```
// Define the pin numbers for the Gas Sensor
const int sensorPin = A0;

void setup() {
  Serial.begin(9600); // Start serial communication at 9600 baud rate
}

void loop() {
  Serial.print("Analog output: ");
  // Read the analog value of the gas sensor and print it to the serial
  // monitor
  Serial.println(analogRead(sensorPin));
  delay(50); // Wait for 50 milliseconds
}
```

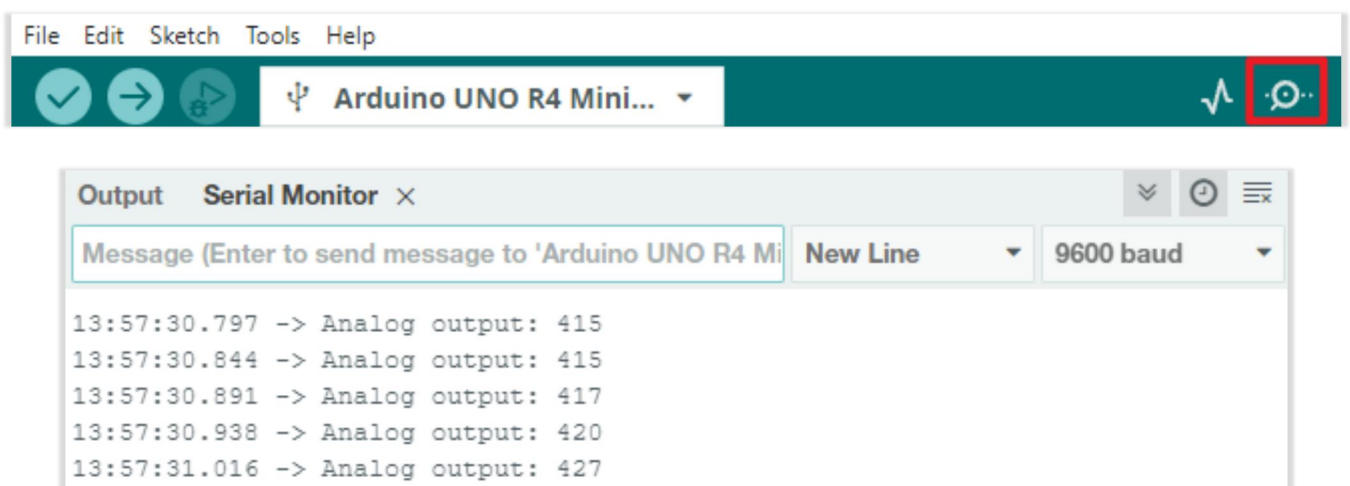
2. Choose the board and port you use.



3. Click the Upload button.



4. Open the **serial monitor** in the upper right corner of the Arduino IDE, set the baud rate to **9600** in the serial monitor, and then you can see the output of the Photoresistor module analog pin. The greater the intensity of light, the smaller the analog output.

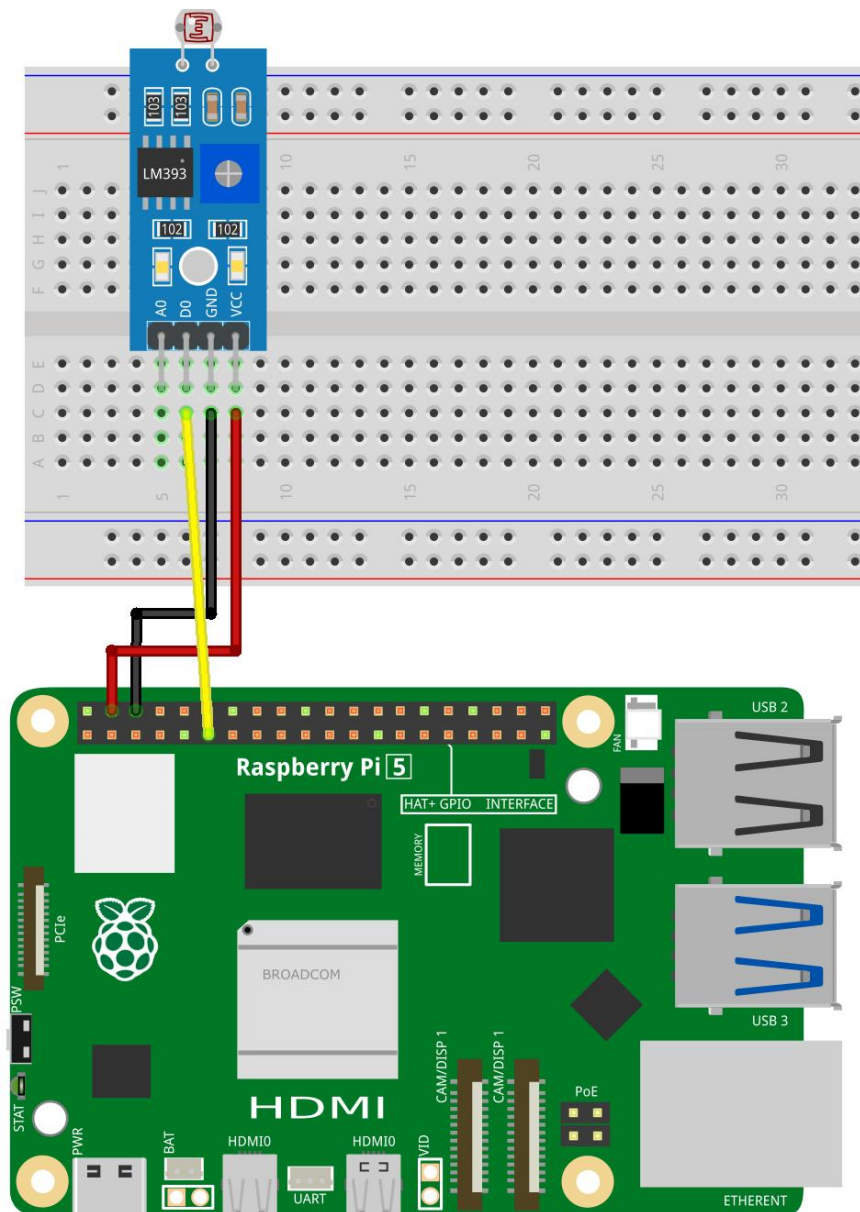


How to Use in Raspberry Pi?

Build the Circuit

Photoresistor Module	Raspberry Pi
VCC	5V
GND	GND
DO	17
AO	

NOTE: You can adjust the light intensity threshold by adjusting the potentiometer on the module.



Copy and Run the Code

1. Copy and paste the following code to your IDE. Save the code below as `photoresistor.py`.

```
from gpiozero import DigitalInputDevice
import time

# Initialize the sensor as a digital input device on GPIO 17
sensor = DigitalInputDevice(17)

# Continuously check the sensor value
while True:
    if sensor.value == 0:
        print("Light detected!") # Print when the sensor detects light
    else:
        print("Darkness detected!") # Print when the sensor detects
darkness

    time.sleep(1) # Wait for 1 second before the next check
```

2. Run the code.

```
python photoresistor.py
```

3. When you run this code, it checks the photoresistor module's status every second. It displays "Light detected!" if light is present, and "Darkness detected!" otherwise. This check repeats continuously until the program is stopped.